ZERO HOUR

DEFUSAL MANUAL

The defuser describes what they see. You find the matching rule and say what to do. Don't look at the bomb.

New to the bomb? Bomb basics, at the end, shows how the case is built and what its parts look like.

Revision 6ebb7e73

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1 WIRES

Cut exactly one wire. Count wires from the top. Use the section that matches the number of wires and take the first rule that applies.

1.1 3 wires

1. If there are no red wires, cut the second wire.
2. Otherwise, if the last wire is white, cut the last wire.
3. Otherwise, if there is more than one blue wire, cut the last blue wire.
4. Otherwise, cut the last wire.

1.2 4 wires

1. If the encoders add up to 12 or more and the last wire is red, cut the last wire.
2. Otherwise, if there is more than one red wire and the last digit of the serial number is odd, cut the last red wire.
3. Otherwise, if the last wire is yellow and there are no red wires, cut the first wire.
4. Otherwise, if there is exactly one blue wire, cut the first wire.
5. Otherwise, if there is more than one yellow wire, cut the last wire.
6. Otherwise, cut the second wire.

1.3 5 wires

1. If the last wire is black and the last digit of the serial number is odd, cut the fourth wire.
2. Otherwise, if there is exactly one red wire and more than one yellow wire, cut the first wire.
3. Otherwise, if there are two coin cells, cut the third wire.
4. Otherwise, if there are no black wires, cut the second wire.
5. Otherwise, cut the first wire.

1.4 6 wires

1. If there are no yellow wires and the last digit of the serial number is odd, cut the third wire.
2. Otherwise, if there is exactly one yellow wire and more than one white wire, cut the fourth wire.
3. Otherwise, if there are no red wires, cut the last wire.
4. Otherwise, if the coin cells add up to 3 volts or more, cut the second wire.
5. Otherwise, cut the fourth wire.

1.5 Rear plate (not on every bomb)

The rear plate is the harness plug the wires run into. It has as many brass pins as the last digit of the serial number: an odd number of pins means the last digit is odd. An empty plug means the digit is 0, which is even. It saves searching the sides of the bomb for the serial.

2 BUTTON

Tell the defuser to hold the button or tap it. Take the first rule that applies.

1. If the button is blue and the label says ABORT, hold it.
2. If there is more than one battery and the label says DETONATE, press and immediately release it.
3. If the button is white and there is a lit indicator labeled CAR, hold it.
4. If there are more than two batteries and there is a lit indicator labeled FRK, press and immediately release it.
5. If the button is yellow, hold it.
6. If there is a coin cell marked CR2016, press and immediately release it.
7. If the button is red and the label says HOLD, press and immediately release it.
8. If the encoders add up to an odd number and the label says PRESS, press and immediately release it.
9. Otherwise, hold it.

2.1 Holding the button

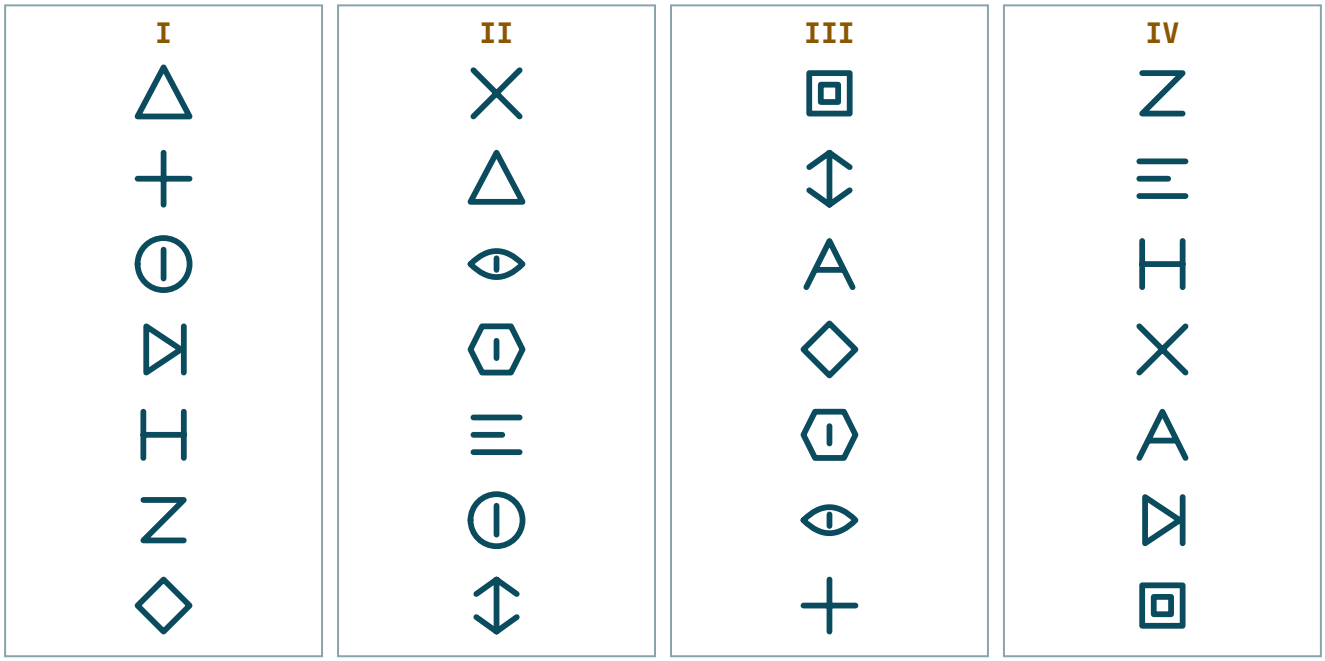
A colored light bar glows beside the button once it is held. Check its color, then tell the defuser to release when the countdown timer shows the matching digit in *any* position.

Light bar color	Release when the timer shows
blue	4
white	1
yellow	5
Any other color	1

"Press and immediately release" means a quick tap, well under half a second.

3 GLYPHS

The four glyphs on the module all appear in exactly one of the columns below. Press them in the order they appear in that column, top to bottom. A wrong press is a strike; glyphs already pressed stay lit.



3.1 Rear plate (not on every bomb)

The rear plate is stamped with four numerals, I to IV. One is raised and amber: that is the column to use, counted left to right in the table above. It saves you searching all four columns.

4 SIMON

The ring flashes a growing sequence of colors. The defuser does *not* repeat the flashes: for each flash, press the color shown in the table instead. The table depends on the serial number and the current number of strikes.

After each correct stage the sequence grows by one flash and replays from the start. Press the translated colors for the whole sequence each time. If the defuser pauses, the sequence repeats. The dots in the center fill in as stages are completed.

4.1 Serial number contains a vowel (A, E, I, O, U)

Strikes	red flashes	blue flashes	green flashes	yellow flashes
0	blue	red	yellow	green
1	yellow	green	blue	red
2	green	red	yellow	blue

4.2 Serial number has no vowel

Strikes	red flashes	blue flashes	green flashes	yellow flashes
0	blue	yellow	green	red
1	red	blue	yellow	green
2	yellow	green	blue	red

4.3 Rear plate (not on every bomb)

The rear plate is the back of the toy: a round speaker grille with a label on the cone. The label lists the vowels in the serial number, or shows a dash when there are none, which picks the table above.

5 PASSWORD

The module shows five columns of letters. Ask the defuser to read out the six letters available in each column (use the arrow keys to cycle through them). Exactly one word from the list below can be spelled using one letter from each column in order. Tell the defuser which word to spell, then press Submit.

about	found	place	still	three
after	great	plant	study	water
again	house	point	their	where
below	large	right	there	which
could	learn	small	these	world
every	never	sound	thing	would
first	other	spell	think	write

6 SWITCHES

Each switch has a lamp above it. A switch should be **up** (tilted toward the top of the module) when its lamp is lit and **down** when it is dark, but some switches are reversed. Work out each switch from the list below, tell the defuser which ones to set up, then have them press Confirm.

1. Switch 1: flip it if the last digit of the serial number is odd.
2. Switch 2: flip it if any indicator is lit.
3. Switch 3: flip it if there are two or more batteries (the AA and D cells, not coin cells).
4. Switch 4: flip it if there is at least one coin cell.
5. Switch 5: flip it if the encoders add up to an odd number. With no encoders the sum is 0, which counts as even.

"Flip it" means the opposite of the lamp: dark means up, lit means down. Rule 5 reads the encoders as they are at the moment of pressing Confirm.

6.1 Rear plate (not on every bomb)

The rear plate is the switch bank's memory backup. Its window reads BACKUP FITTED when the bomb has any coin cells and BACKUP EMPTY when it has none, which is what switch 4 needs. The window is a printed flag, not a coin cell: don't count it as one.

7 MORSE

The lamp flashes a word in Morse code, then pauses and repeats. A short flash is a dot, a long flash is a dash, and a longer dark gap separates letters. Decode the word with the chart, find its frequency in the table, tell the defuser to tune to it with the arrow keys and press Transmit.

A ·–	**B** –···	**C** –·–·	**D** –··
E ·	**F** ··–·	**G** ––·	**H** ····
I ··	**J** ·–––	**K** –·–	**L** ·–··
M ––	**N** –·	**O** –––	**P** ·––·
Q ––·–	**R** ·–·	**S** ···	**T** –
U ··–	**V** ···–	**W** ·––	**X** –··–
Y –·––	**Z** ––··		

Word	Frequency (MHz)
fuse	3.505
volt	3.515
coil	3.522
bolt	3.532
grid	3.535
relay	3.542
laser	3.545
diode	3.552

8 MEMORY

Five stages. Each stage the display shows a number from 1 to 4 and the four keys carry the labels 1 to 4 in a new order. Positions are counted left to right. Look up the stage and the displayed number below and tell the defuser which key to press. **Write down the position and the label of every key pressed**, because later stages refer back to them. A wrong press is a strike and sends the module back to stage 1.

Stage	Shows 1	Shows 2	Shows 3	Shows 4
1	press the key in position 2	press the key in position 1	press the key in position 4	press the key in position 3
2	press the key labeled 3	press the key in the same position as in stage 1	press the key in position 1	press the key with the same label as in stage 1
3	press the key with the same label as in stage 2	press the key in position 4	press the key in the same position as in stage 1	press the key labeled 1
4	press the key in position 3	press the key with the same label as in stage 3	press the key in the same position as in stage 2	press the key labeled 2
5	press the key in the same position as in stage 4	press the key with the same label as in stage 1	press the key with the same label as in stage 4	press the key in the same position as in stage 3

9 KNOB

Needy. This module is never defused. Every so often its six lamps (two rows of three) light up in a pattern and its countdown starts. Find the pattern in the table to learn where the red notch of the knob must point, then tell the defuser to click the knob round (each click turns it a quarter clockwise) and press Lock before the countdown runs out. Filled circles are lit lamps. "Up" is the top of the module.

Pattern	Notch points
● ○ ○ ● ● ○	up
○ ● ● ○ ○ ●	up
● ● ○ ○ ● ●	right
○ ○ ● ● ○ ○	right
● ○ ● ○ ● ○	down
○ ● ○ ● ○ ●	down
● ● ● ○ ○ ○	left
○ ○ ○ ● ● ●	left

Locking the wrong way is a strike, and so is a countdown that runs out. Either way the lamps go dark until the next time.

10 BREAKER

Needy. This module is never defused: every so often it trips, showing a new number and lamp, and its countdown starts. It is linked to the double-throw knife switch on the left or right panel of the bomb: silver blades on a centre hinge with a ribbed black handle. Thrown up (blades in the top contacts) is ON, and thrown a full half turn down (blades in the bottom contacts) is OFF. A green lamp at the top and a red lamp at the bottom show which. The defuser must put the lever in the right position and then press Latch. Work out the position from the displayed number and the lamp:

1. Add the two digits of the displayed number. If the sum is even the switch should be ON (thrown up). If it is odd it should be OFF (thrown down).
2. If the last digit of the serial number is odd, swap: ON becomes OFF and OFF becomes ON.
3. If the lamp is red, swap again. A green lamp changes nothing.

Drag the handle up or down (or just click it) to throw the switch, then press Latch before the countdown runs out. A wrong Latch is a strike, and so is a countdown that runs out. Either way the breaker rests until it trips again.

10.1 Rear plate (not on every bomb)

The rear plate is the breaker's rating plate. Its trip coil number is the last digit of the serial number; rule 2 needs to know whether it is odd.

11 COMBINATION

The three digits on the display are a code for the rotary encoders engraved **A**, **B** and **C** on the side panels of the bomb. Each encoder has to be turned to the reading below, then the defuser presses Verify.

1. Encoder A: the first digit plus the number of lit indicators.
2. Encoder B: the second digit plus the number of batteries (the AA and D cells, not coin cells).
3. Encoder C: the third digit plus the number of coin cells.

Only the last digit of each sum counts, so 13 becomes 3. To set an encoder, click the right half of its knob to raise the reading by one and the left half to lower it (9 wraps to 0).

12 FUSES

Five fuses sit side by side. Pull exactly one: the one that feeds the circuit named by the first rule that applies. A wrong fuse is a strike.

1. If the serial number contains a vowel, pull the fuse that feeds RADIO.
2. Otherwise, if there are two or more batteries, pull the fuse that feeds SPARK.
3. Otherwise, if any indicator is lit, pull the fuse that feeds LAMPS.
4. Otherwise, pull the fuse that feeds CLOCK.

Which fuse feeds which circuit is only written on the back, so the defuser has to turn the bomb. The amp ratings on the fuses don't matter.

12.1 Rear plate

The rear plate is the fuse block's wiring: five terminals, each labelled with its circuit. Seen from behind, the labels run the other way: the first label from the right on the back belongs to the first fuse from the left on the front.

13 GEARS

The defuser turns a crank on the front, with the two buttons (one full turn per press) or by dragging it round, then presses Release. A counter on the front counts clockwise turns up and anticlockwise turns down. The front also shows the load: **LIFT** or **DROP**.

1. Direction. LIFT needs the winch drum to turn clockwise, DROP anticlockwise, both as seen from the front. Neighbouring gears turn opposite ways, so count every gear on the back, from the crank's gear to the drum's gear: with an odd number the drum turns the same way as the crank, with an even number the opposite way.

2. Turns. The crank must turn as many times as there are gears with a red dot.

So press Release when the counter shows the number of red-dot gears: as it is if the crank must turn clockwise, with a minus sign if it must turn anticlockwise. Turning the wrong way only moves the counter; turn back to fix it. Release at any other number is a strike and sets the counter back to 0.

13.1 Rear plate

The rear plate is the gear train: a chain of three to six gears from the crank to the winch drum, some with a red dot. The drawing does not move, so count the gears rather than watch them. Seen from behind the chain runs the other way; that does not change the count.

14 RELAY LOGIC

Three jumpers, A, B and C, each set to 0 or 1, feed a relay board whose schematic is on the back: gate 1 takes A and B, gate 2 takes gate 1's output and C, and gate 2 drives the output. The front shows the target, ON (output 1) or OFF (output 0).

Have the defuser describe the two gate symbols on the back, look them up below, and find the one jumper setting that gives the target. Then press Test. Any other setting is a strike.

A small circle on an input line, before it reaches a gate, is an inverter: it flips that input (0 becomes 1, 1 becomes 0). Ask the defuser whether A, B or C has one.



AND
1 only when both inputs are 1.



OR
1 when either input is 1.



NAND
0 only when both inputs are 1.



NOR
1 only when both inputs are 0.

The circle at the output of NAND and NOR is part of the symbol: it flips the gate's output.

14.1 Rear plate

The rear plate is the relay board's schematic: inputs A, B and C on the left, some with an inverter circle on their line, two gates drawn as symbols without names, and the output on the right. Only one setting of the three jumpers gives the target.

15 COOLANT

Needy. This module is never defused. While it idles, its green IDLE lamp is lit and the gauge rests at 0. Every so often it wakes: the IDLE lamp goes out, a new number appears on the demand display, the valve on the back slips to another setting, and its countdown starts. Work out the target pressure:

1. Start with the number on the demand display.
2. Add the number of lit indicators.
3. Subtract the number of batteries (the AA and D cells, not coin cells).
4. If the result is below 1, use 1; above 9, use 9.
5. The rear plate carries a tag, RELIEF *n* BAR. If the result is above *n*, use *n* minus 1 instead.

Have the defuser set the valve on the back to the target, then turn the bomb and watch the gauge on the front: the needle moves about one bar per second towards the valve setting. Press Seal when it

points at the target, before the countdown runs out. Sealing at any other pressure is a strike, and so is a countdown that runs out; either way the module idles again until its next wake.

15.1 Rear plate

The rear plate is the coolant valve: a red hand wheel on a scale from 0 to 9, and the relief tag. Clicking the right half of the wheel opens the valve one step, the left half closes it. The pointer on the wheel shows the setting.

16 THROTTLE

A lever with positions 1 (bottom) to 5 (top) and a coloured grip. Set the lever, then press Engage. Engaging at the wrong position is a strike.

Grip colour	Position
red	2
blue	4
yellow	1
white	5
black	3

If there are more than two batteries (the AA and D cells, not coin cells), use one position lower, but never below 1.

17 MAZE

The module shows a 6×6 grid with two green rings, the defuser's white square and a red goal. The walls are invisible. Guide the square to the goal with the four arrows; walking into a wall is a strike and the square stays put.

1. Ask for the gyro plate on the back. It reads ROTATED 0°, 90° CW, 180° or 270° CW: the maze is mounted turned that far clockwise, as seen from the front.
2. Turn this page the same way (clockwise by that angle). Now the mazes below look exactly like the module.
3. Find the maze whose two green rings sit where the defuser sees them, then talk the square to the goal: up, down, left and right are the defuser's.

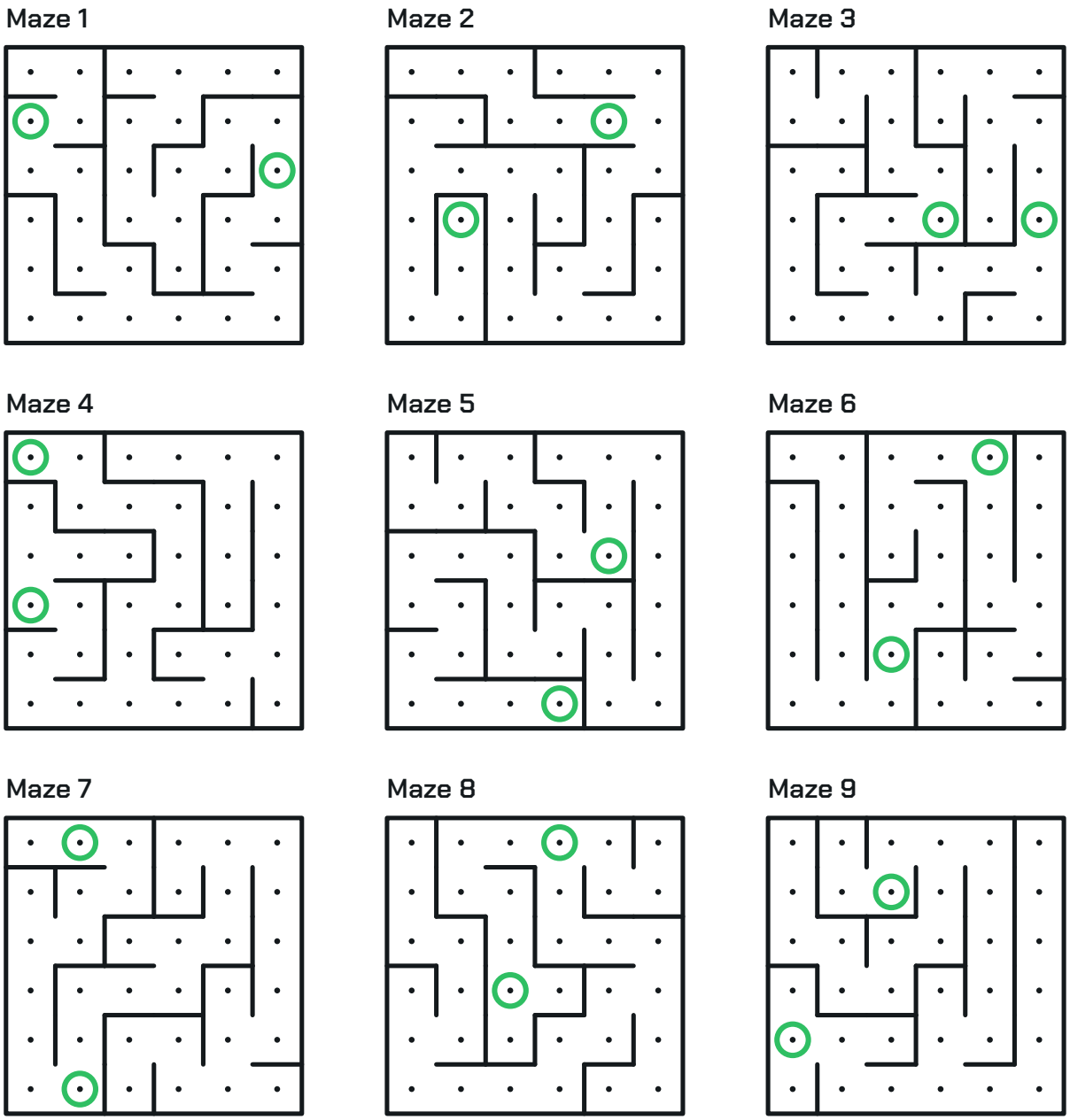


Figure 1. The nine mazes, drawn unturned. Green rings mark how to tell them apart.

Gyro	Defuser's ring at column c, row r is on the unturned map at
0°	column c, row r
90° CW	column r, row 7 – c
180°	column 7 – c, row 7 – r
270° CW	column 7 – r, row c

Columns count from the left and rows from the top, 1 to 6, as the defuser sees them.

17.1 Rear plate

The rear plate is the gyro mount. Its stamped text says how far the maze is turned clockwise, seen from the front. Read the text, not the plate's orientation: seen from behind, left and right swap.

18 CIPHER

The display shows five scrambled letters. Each one was moved forward through the alphabet by the digit of its rotor on the back of the bomb (Z wraps round to A). Move each letter back by its digit to get the word, have the defuser dial it in on the five wheels and press Submit. A wrong word is a strike.

The rotors are on the back, so their order is mirrored: the first digit from the right on the back belongs to the first letter on the front.

A 1	B 2	C 3	D 4	E 5	F 6	G 7	H 8	I 9	J 10	K 11	L 12	M 13
N 14	O 15	P 16	Q 17	R 18	S 19	T 20	U 21	V 22	W 23	X 24	Y 25	Z 26

The word is always one of these:

alarm	ember	laser	quake	vault
blast	flare	nerve	radar	watch
cable	ghost	omega	shock	delta
havoc	panic	timer		

18.1 Rear plate

The rear plate is the rotor bank: five digits from 1 to 9. Seen from behind, they run the other way round: the rightmost digit belongs to the first letter.

19 RESISTORS

Three resistors, R1 to R3, each with three colour bands read from the left: two digits, then a multiplier. Ignore the silver band at the right end. Black multiplies by 1, brown by 10. The wiring is drawn on the back of the bomb. Work out the total resistance in ohms, have the defuser dial it in on the three wheels (hundreds, tens, ones) and press Measure. A wrong total is a strike.

Colour	Digit	As multiplier
black	0	×1
brown	1	×10
red	2	
orange	3	
yellow	4	
green	5	
blue	6	
violet	7	
grey	8	
white	9	

1. All three in a row (in series): add them up. R1 + R2 + R3.

2. R1, then R2 and R3 side by side (in parallel): R2 and R3 are always equal, and together they count half of one. R1 + R2 ÷ 2.

Example: red, red, brown is 22 × 10 = 220 ohms.

19.1 Rear plate

The rear plate is the wiring diagram: either R1, R2 and R3 in one row, or R1 followed by R2 and R3 side by side.

20 SIGNAL

An oscilloscope shows one of four waveforms, repeated one to four times. Ask the defuser for its shape and how many peaks (highest points) it has, then work out the three knob settings below. Each click on a knob steps it; Lock checks all three. Any wrong setting is a strike.

Shape	Looks like	AMP
sine	smooth waves	3
square	flat tops and bottoms with straight jumps	5
saw	falls in a straight line, then jumps back up	2
triangle	straight lines up and down to sharp points	4

1. FREQ: the number of peaks plus the number of lit indicators. If that is above 5, use 5.
2. AMP: from the table above.
3. PHASE: from the table below.

Last digit of the serial number	No coin cell	At least one coin cell
even	0	90
odd	180	270

21 MULTIMETER

Six jacks, A to F, each at its own voltage, a red and a black probe and a needle meter. The defuser clicks a jack to put a probe on it: the first click places the red probe, the second the black one, and each further click moves the probe that has stood longest. The meter shows the voltage between the two probes. When the probes are on the right pair, the defuser presses Patch to plug a cable across them.

1. The target is the total voltage of the coin cells (a CR2032, CR2025 or CR2016 is 3 V, an LR44 1.5 V).
2. With no coin cells, the target is 1.5 V for every battery (the AA and D cells).
3. With no coin cells and no batteries, the target is 12 V.

Exactly one pair of jacks reads the target. Every other pair reads at least 1 V away from it, and the scale has a tick every half volt. Patching any other pair is a strike.

21.1 Rear plate (not on every bomb)

The rear plate is the power supply label: the number of batteries (cells) and the total voltage of the coin cells.

22 SEMAPHORE

A signaller shows four letters in flag semaphore, one after another; the defuser steps through them with the arrows. Three of the letters are in the serial number, one is not. Have the defuser press ACK while that one is showing. Acknowledging any other letter is a strike.

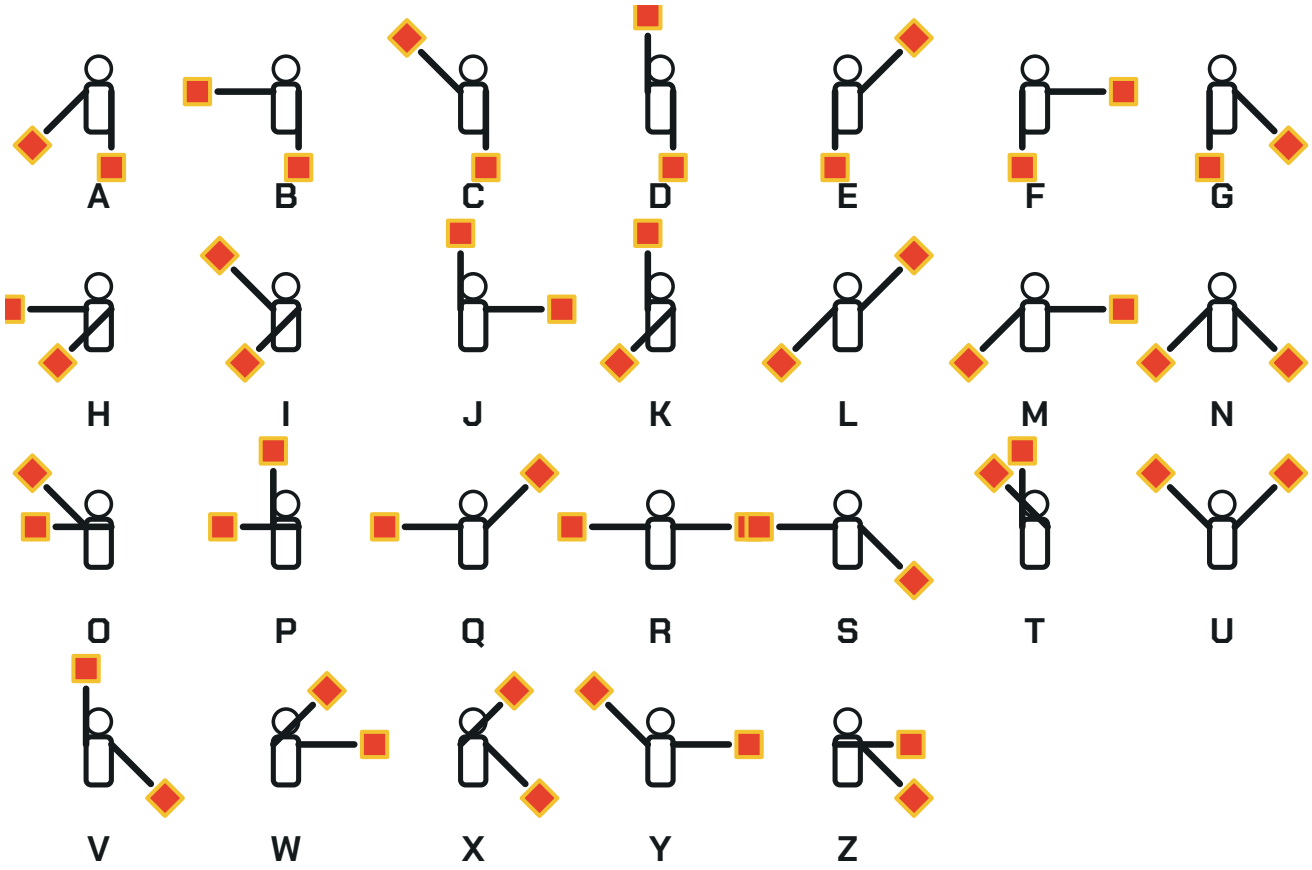


Figure 2. The semaphore letters, as seen from the front. The flags are red and yellow.

Some letters look alike from a distance: describe both arms, for example "one arm straight down, the other out to the left".

22.1 Rear plate (not on every bomb)

The rear plate is the call-sign plate: the letters of the serial number, so the defuser does not have to search the case for it.

23 DIP SWITCHES

Four slide switches, numbered 1 to 4 from the left. Add up the number of batteries (the AA and D cells, not coin cells), the number of lit indicators and the last digit of the serial number. If the total is 16 or more, take away 16. Look it up below, have the defuser set the switches (▲ up, ▼ down) and press Apply. A wrong pattern is a strike.

Total	Switches 1 2 3 4	Total	Switches 1 2 3 4
0	▼▼▼▼	8	▲▼▼▼
1	▼▼▼▲	9	▲▼▼▲
2	▼▼▲▼	10	▲▼▲▼
3	▼▼▲▲	11	▲▼▲▲
4	▼▲▼▼	12	▲▲▼▼
5	▼▲▼▲	13	▲▲▼▲
6	▼▲▲▼	14	▲▲▲▼
7	▼▲▲▲	15	▲▲▲▲

23.1 Rear plate (not on every bomb)

The rear plate is stamped with the code base: the number of batteries plus the number of lit indicators. Only the serial number's last digit is left to add.

A BOMB BASICS

A.1 The case

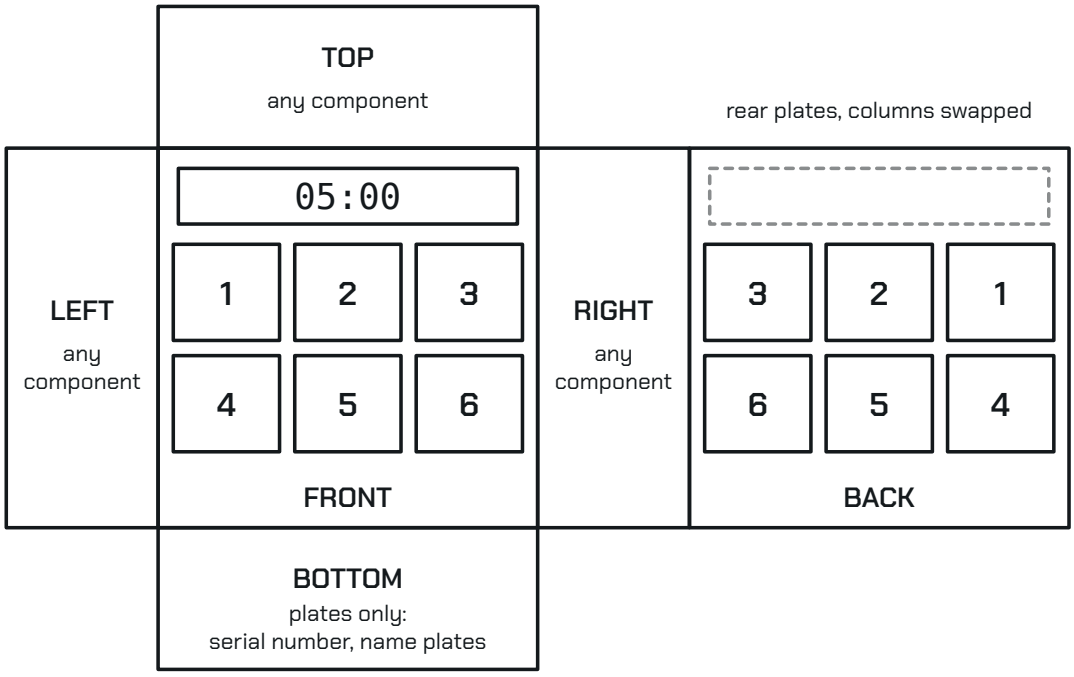


Figure 3. The case unfolded. Modules sit in the front bays. A rear plate sits directly behind its module, so seen from behind the columns swap: the plate for module 1 is on the right.

- **Rotate the case.** Modules sit on the front. Everything else is on the other sides, and the positions change on every bomb. The left, right and top panels can carry any component. The bottom only ever carries plates: the serial number plate, and sometimes plain name plates (model or rating) that mean nothing. If a rule needs a component, ask the defuser to turn the bomb and look.

- **Rear plates.** Some modules have a steel rear plate on the back of the bomb, directly behind the module. Seen from behind, left and right swap: a module in the left column on the front has its plate in the right column on the back. Some rear plates repeat a clue from elsewhere on the bomb; others are part of the puzzle, and their module cannot be solved without them. A module without a rear plate has a blank panel behind it. Each module's section says what its plate shows.

A.2 Parts on the side panels

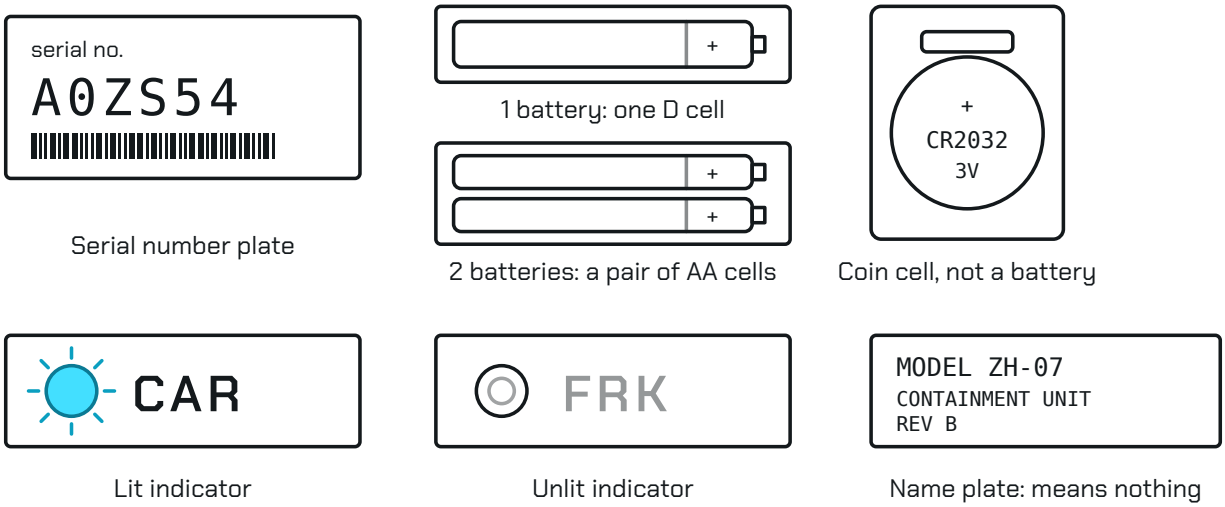


Figure 4. Parts on the side panels: the serial number plate, battery holders, a coin cell, indicators, and a plain name plate.

- **Serial number.** A white steel plate with a barcode. Rules refer to the last digit (odd or even) and to whether the serial contains a vowel: A, E, I, O or U.
- **Batteries.** Count the individual cells in all battery holders (a single large D cell, or a pair of AA cells). There may be none at all. Coin cells are not batteries.
- **Coin cells.** Flat silver discs in small black holders with a steel clip. The model is stamped on top: CR2032, CR2025 and CR2016 are 3 volt lithium cells (in that order they get thinner), and LR44 is a small, thick 1.5 volt alkaline cell. A bomb can have none, one or two.
- **Indicators.** Small labeled plates with a lens. A lit indicator has a glowing lens and bright text. An unlit one is dark.

A.3 Rotary encoders

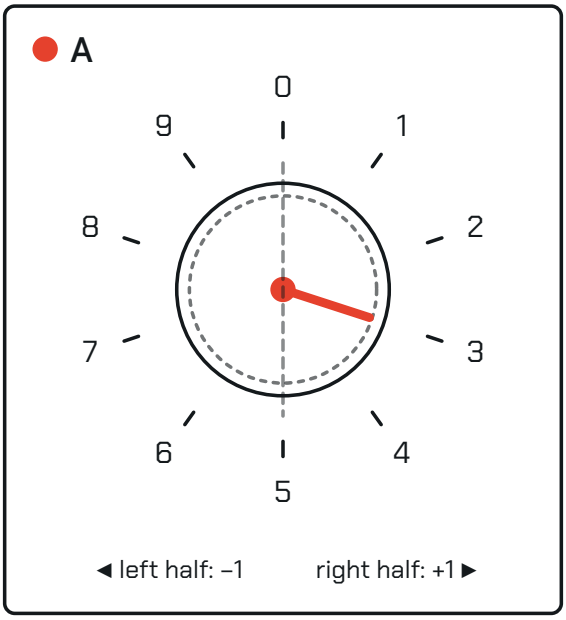


Figure 5. Rotary encoder A, reading 3: the red pointer aims at the reading. Clicking the right half of the knob raises the reading by one, the left half lowers it.

- **Rotary encoders.** Some bombs have up to three: a knurled aluminium knob on a dial numbered 0 to 9, with a letter (A, B, C) and a colored cap. The reading is the digit the red pointer line aims at, and the digits are printed around the knob. Click the right half of a knob to turn it one step clockwise (the reading goes up, 9 wraps to 0) or the left half to turn it back one step. The mouse wheel and right-click also turn it. Rules read the encoders as they are at the moment you act, so turning one changes the answer. "The encoders add up" means the sum of all readings, and a bomb without encoders adds up to 0. A bomb can carry encoders that no rule uses.

A.4 Knife switch

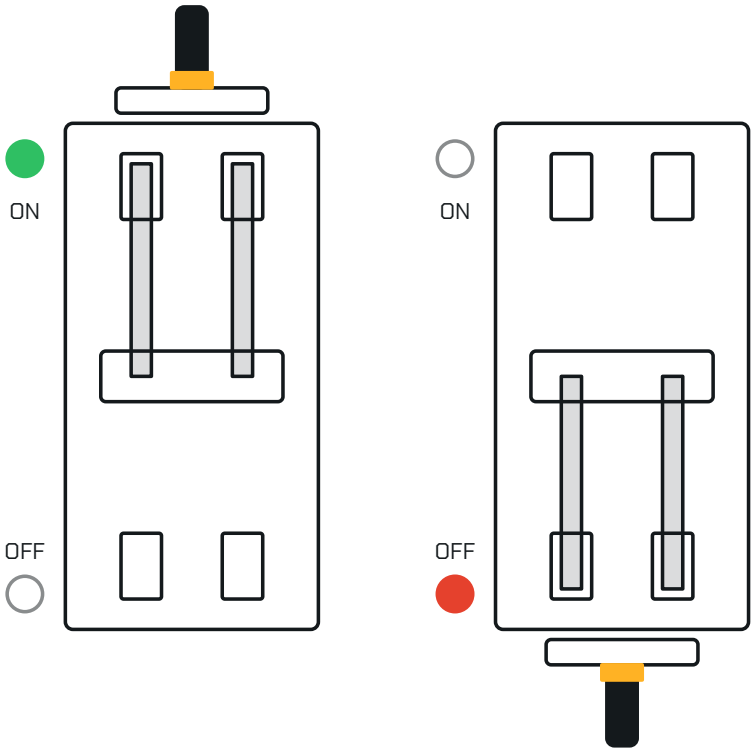
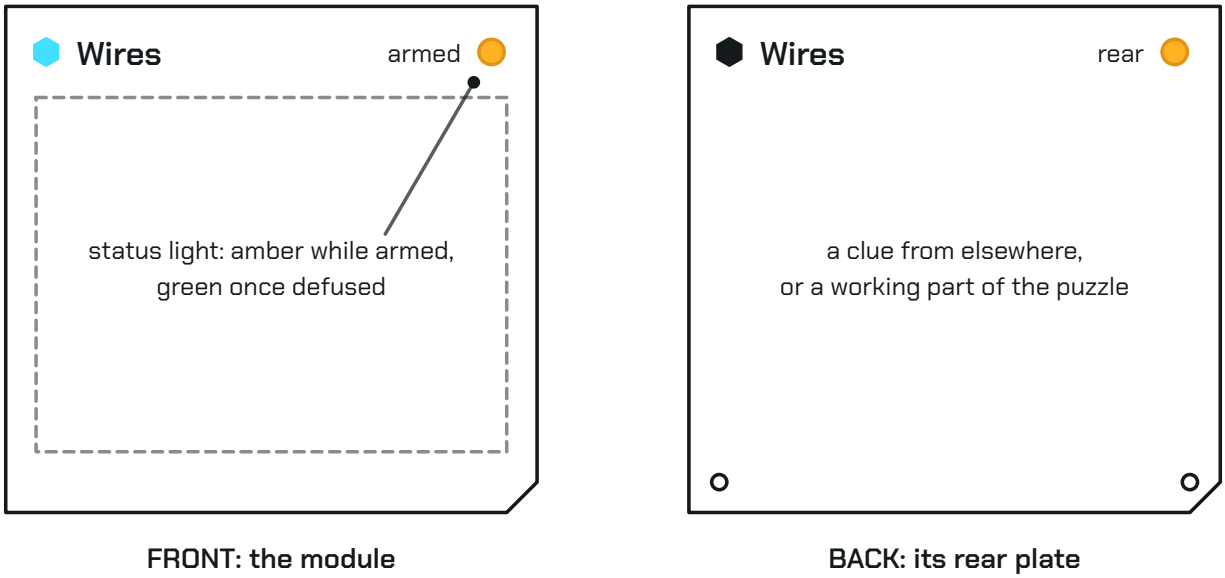


Figure 6. The knife switch. ON: thrown up, blades in the top contacts, handle above the base, green lamp lit. OFF: thrown a half turn down, blades in the bottom contacts, handle below the base, red lamp lit.

- **Knife switch.** Some bombs have a double-throw knife switch on the left or right panel: a dark graphite base with two silver blades on a centre hinge, silver contacts and terminal blocks with socket-head screws at the top and the bottom, and a ribbed black handle with an orange collar. Thrown up, with the blades in the top contacts and the handle above the base, it is ON. Thrown a full half turn down, with the blades in the bottom contacts and the handle below the base, it is OFF. A green lamp at the top and a red lamp at the bottom show which. Drag the handle up or down, or click it to throw it. Only the Breaker module reads it; on any other bomb it is a decoy.

A.5 Modules and strikes



FRONT: the module **BACK: its rear plate**

Figure 7. A module on the front and its rear plate on the back. Some modules have no rear plate; then the panel behind them is blank.

- **Modules.** Each module is independent. Defuse them all before the clock reaches zero. A module's status light turns green when it is safe.
- **Needy modules.** Some modules are never defused. They sleep, then wake with a small countdown in their corner and need attention before it runs out, again and again. Their plate says *needy* instead of armed. The bomb is defused once every other module is safe.
- **Strikes.** Every mistake is a strike, and each strike makes the clock run 25% faster. The third strike detonates the bomb.